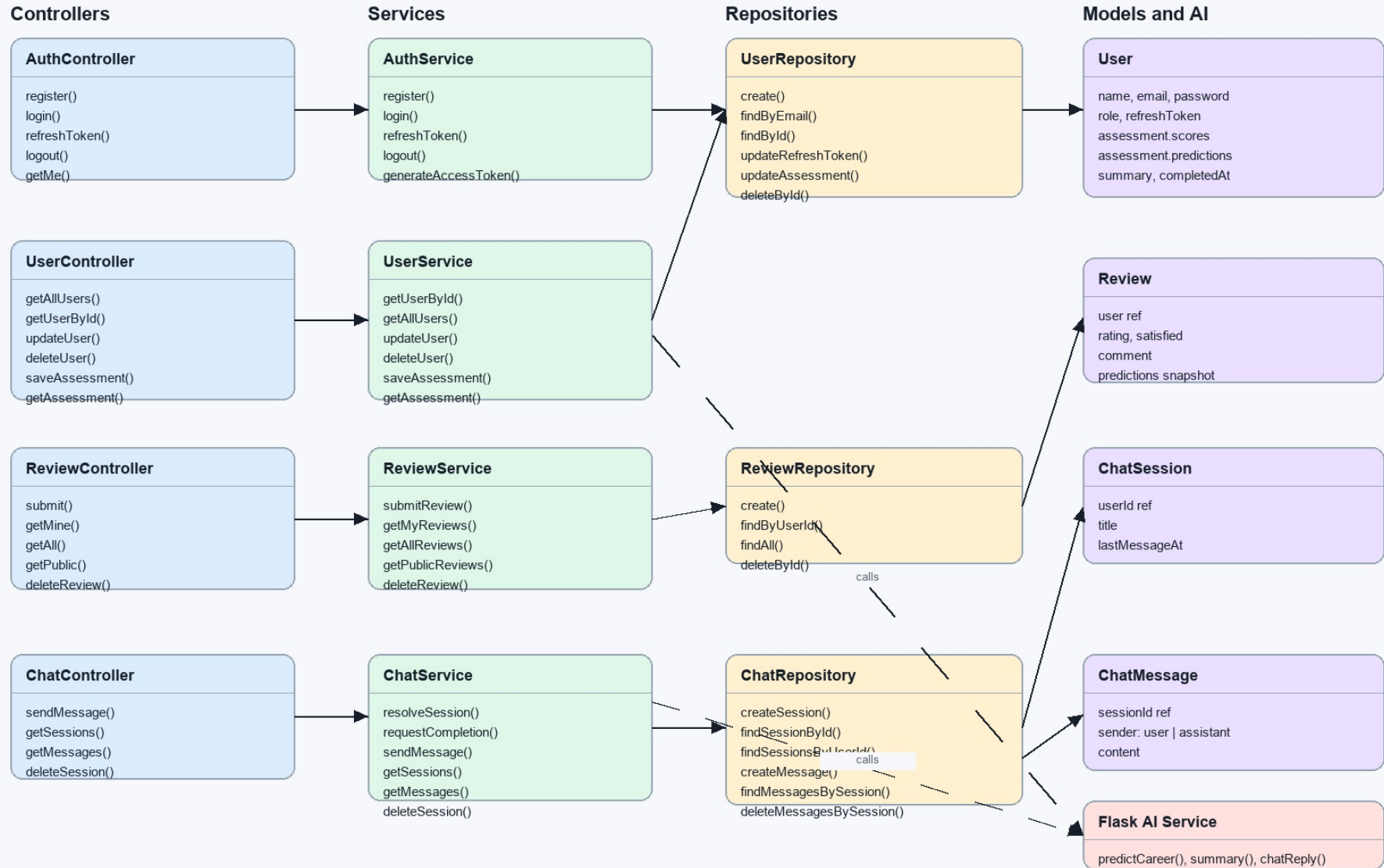


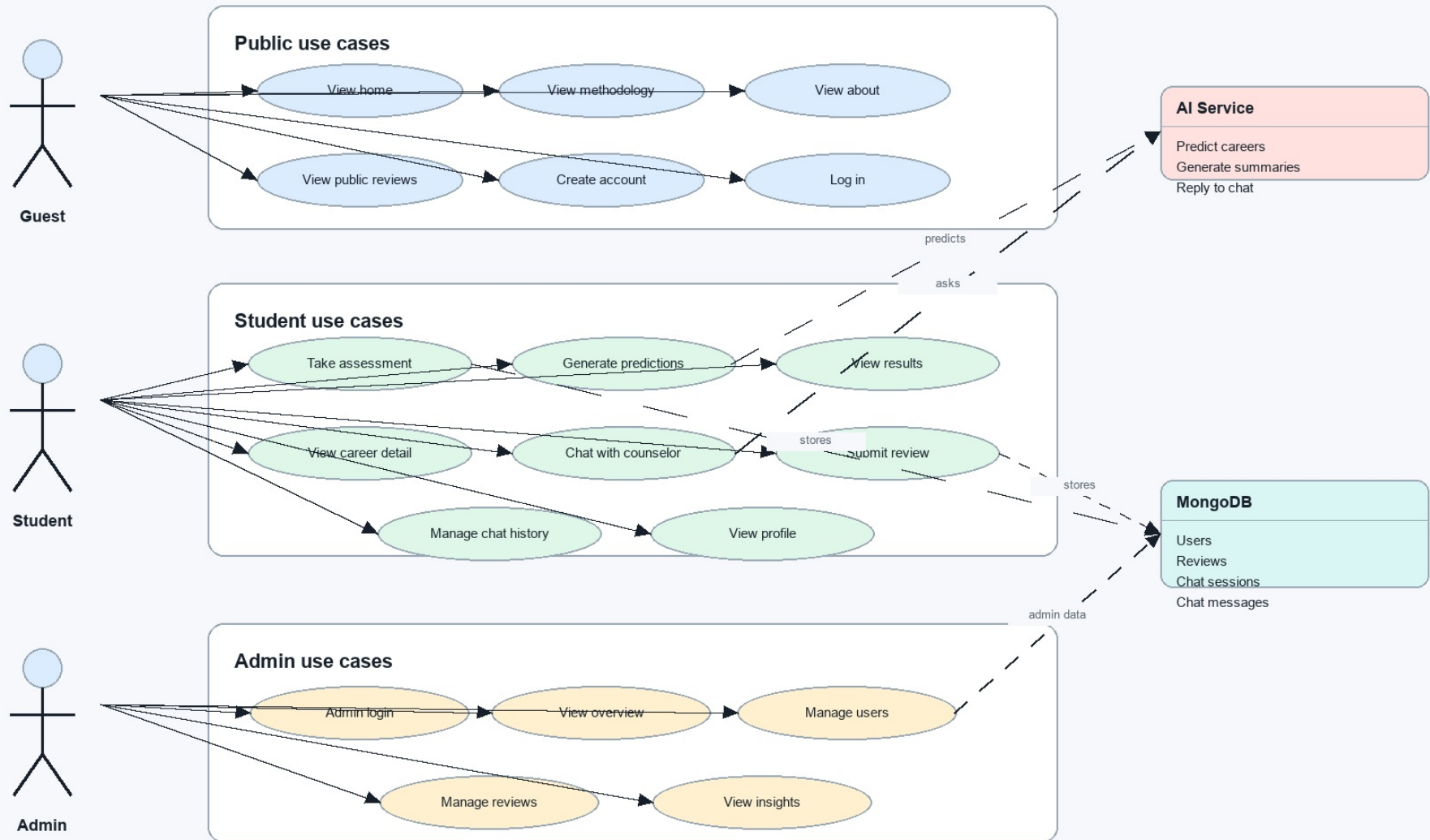
Backend Class Diagram

Express controllers delegate to services, repositories, Mongoose models, and the Flask AI service.



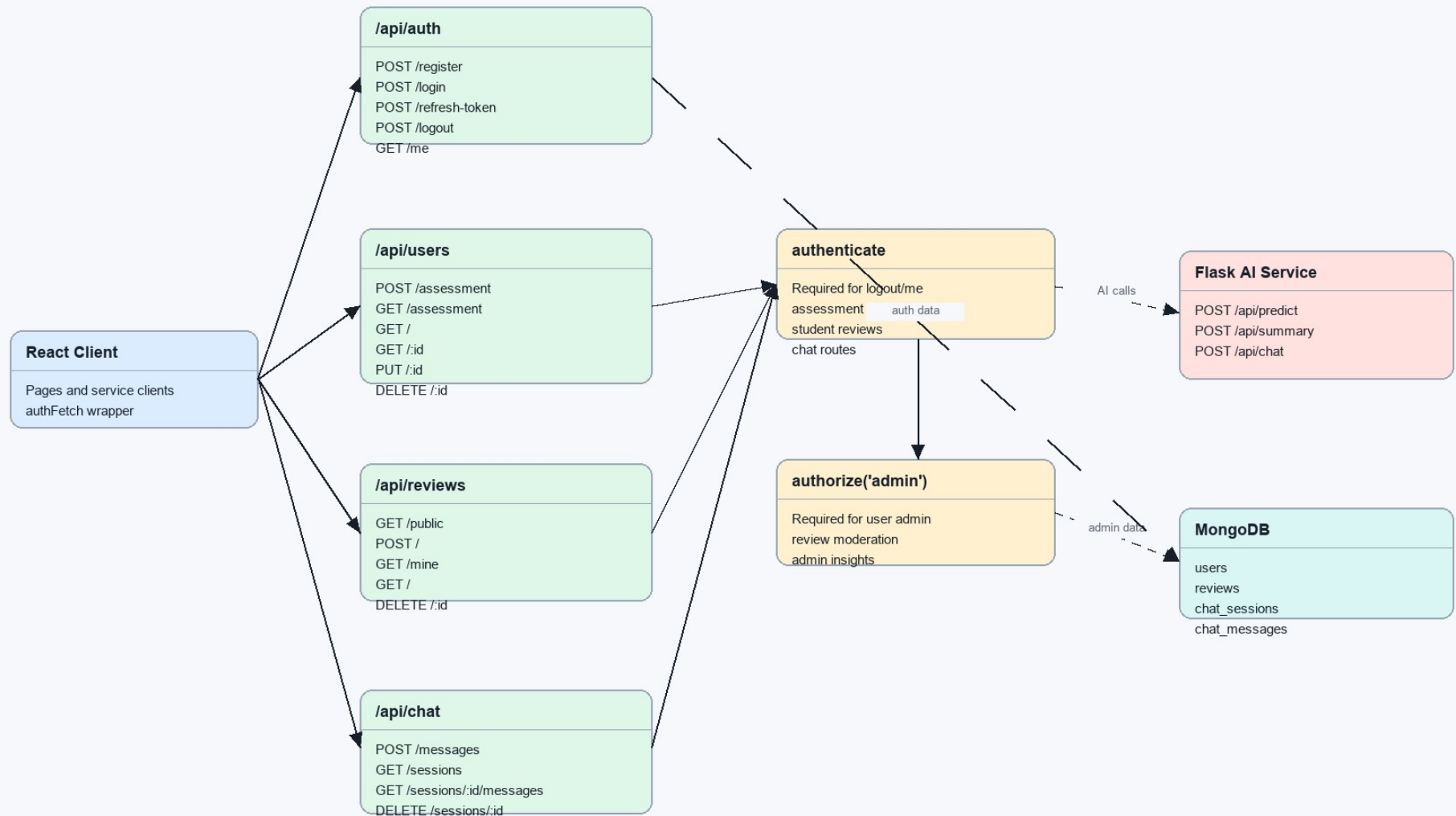
Use Case Diagram

Primary user goals for guests, registered students, admins, and the AI service.



API Use Case Diagram

React client route groups, authentication middleware, admin authorization, and AI service calls.



MongoDB Collection Schema Diagram

Collections, embedded documents, references, and cardinality in the Mongoose data model.

